

Dr. Jan Friedrich

Curriculum Vitae

Technical University of Munich
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Personal Data

Nationality German
Homepage <https://janfrie-dev.github.io/home/>

Working Experience in Academia

- since 09/2025 **Researcher/Lecturer (Akademischer Rat auf Zeit)**, TU Munich, School of Computation, Information and Technology Department of Mathematics, Chair of Optimal Control, Prof. Dr. Boris Vexler
- Research related to Numerical Analysis, Optimization and Control of partial differential equations
 - Exercise classes and courses in *Nonlinear Optimization*
 - Digitalization of academic administration through a tool for exam supervision and grading allocation
- since 09/2023 **Principal Investigator**, German Research Foundation, SPP 2410: Hyperbolic Balance Laws in Fluid Mechanics: Complexity, Scales, Randomness
Project: Balance laws with space-dependent nonlocalities: modeling, simulation and uncertainty quantification
- Analysis of multidimensional nonlocal partial differential equations
 - Numerical methods for multidimensional systems of nonlocal balance laws
 - Stochastic influences on the nonlocal dynamics
 - Co-supervision of Anika Beckers (PhD student at RWTH Aachen University)
- 01/2022–08/2025 **Postdoctoral Researcher**, RWTH Aachen University, Institute of Geometry and Applied Mathematics, Prof. Dr. Michael Herty
- Research related to Numerical Analysis, Optimization and Control of partial differential equations, including inverse problems
 - Lecturer of *Applications of scalar conservation laws*
- 09/2021–12/2021 **Postdoctoral Researcher**, University of Mannheim, Chair of Scientific Computing, Prof. Dr. Simone Göttlich
- Lecturer of *Numerical methods for ordinary differential equations*
- 09/2017–08/2021 **Research Assistant**, University of Mannheim, Chair of Scientific Computing, Prof. Dr. Simone Göttlich
- Research on Numerical Analysis of nonlocal partial differential equations
- 02/2016–07/2016 **Student Assistant**, University of Mannheim, Chair of Scientific Computing, Prof. Dr. Oliver Kolb
- Tutor for Numerics
- 08/2015–01/2016 **Student Assistant**, University of Mannheim, Chair of Stochastic, Prof. Dr. Leif Döring
- Tutor for Analysis 1
- 02/2015–07/2015 **Student Assistant**, University of Mannheim, Chair of Business Mathematics II, Prof. Dr. Andreas Neuenkirch
- Tutor for Numerics

Working Experience outside Academia

- 10/2016–02/2017 **Horváth & Partners Management Consultants, Steering Lab**, Munich, Internship
- Analysis of price elasticities
 - dynamic pricing and prize optimization in an offline-store using receipt data
- 07/2016–09/2016 **Deutsche Bank, Risk (Rating Methodologies)**, Frankfurt, Internship
- Re-calibration and optimization of the *Leveraged and Structured Finance Scorecard*
 - Regression-models using macro-economic data within the IFRS9 project

Education

- 09/2017–08/2021 **PhD in Mathematics (Dr. rer. nat.)**, *Chair of Scientific Computing*, University of Mannheim, Prof. Dr. Simone Göttlich
- Thesis: Traffic flow models with nonlocal velocity
- Grade: summa cum laude
- Committee: Prof. Dr. Simone Göttlich, Dr. Paola Goatin, Prof. Dr. Axel Klar
- 02/2015–08/2017 **Master Studies, Business Mathematics**, University of Mannheim
- Degree: *Master of Science with distinction*, Final Grade: 1.2
- 09/2013–01/2014 **Semester abroad**, *Swansea University*, Swansea, Wales
- 09/2011–01/2015 **Bachelor Studies, Business Mathematics**, University of Mannheim
- Degree: *Bachelor of Science*, Final Grade: 1.3
- 2011 **High school diploma (Abitur)**, *Albert-Schweizer-/Geschwister-Scholl Gymnasium*, Marl, Final Grade: 1.4

Research Stays

- 02/2025 **Indian Institute of Petroleum and Energy (IIPE)**, *Visakhapatnam, India*, Paired Early Career Fellowship in Applied Research with Dr. Rathan Samala, financially supported by the Indo-German Science and Technology Centre (IGSTC)
- 02/2024 **Indian Institute of Petroleum and Energy (IIPE)**, *Visakhapatnam, India*, Dr. Rathan Samala, financially supported by IIPE
- Lecturer of a Ph.D. course on *Applications of scalar hyperbolic conservation laws*
- 08/2019 **Arizona State University**, *Phoenix, USA*, Prof. Dr. Dieter Armbruster, financially supported by DAAD-PPP USA
- 07 and 11/2019 **INRIA**, *Sophia Antipolis, France*, Dr. Paola Goatin, financially supported by DAAD France and PHC Procope
- 01/2019 **INRIA**, *Sophia Antipolis, France*, Research Tandem jointly with Dr. Felisia Angela Chiarello, financially supported by DAAD and BMBF

Third-party funds and grants received

- since 08/2025 **Exploratory Research Space**, RWTH Aachen University, internal project funding
- Project: Regulation and Stabilization of Stem Cell Dynamics in Blood Cancer and Bone Marrow Failure Patients, together Dr. Wenhui Shi and Prof. Dr. Thomas Stiehl
 - own/total Budget: 16.700 €/49.000 €
 - Due to the change to TU Munich the funding remains at RWTH Aachen University
- 07/2024 **Paired Early Career Fellowship in Applied Research (PECFAR)** by the Indo-German Science and Technology Centre (IGSTC)
- Research tandem with Dr. Rathan Samala: visits in India 02/2025 and in Germany 05-07/2025
 - own/total Budget for Travel: 3,200 €/9,000 €

- 02/2024 **Indian Institute of Petroleum and Energy**, Funding for a research stay: $\approx 3,500$ €
 since 09/2023 **Principal Investigator**, DFG-SPP 2410: Hyperbolic Balance Laws in Fluid Mechanics: Complexity, Scales, Randomness
Project: Balance laws with space-dependent nonlocalities: modeling, simulation and uncertainty quantification (own/total Budget)
 ○ Staff: 84,800 € / 169,600 € (two 75%-Positions for 18 months each)
 Travel and Guests: 9,000 € / 18,000 €
 Programme allowance for indirect project costs: 20,636 € / 41,272 €
 ○ *Due to the change to TU Munich the funding remains at RWTH Aachen University*
- 09/2023 **Organization of two workshops** on *Recent Trends in Optimization and Control* in Pretoria, South Africa and Dakar, Senegal, financed by the Volkswagen Foundation, Grant: 94,000 €
- 06/2020 **IDEUM**, Funding for a conference trip, Grant: 1,000 €
- 01/2019 **IPID4all mobility grant**, Research Tandem financially supported by DAAD and BMBF, Grant: 1.468 €

Academic self-administration

- since 09/2025 Digitalization of academic administration through a tool for exam supervision and grading allocation, Technical University of Munich
- 08/2023-07/2024 Member of the Search Committee W3 *Analysis and its Applications*, RWTH Aachen University
- 08/2022-07/2024 Deputy member of the Examination Board for Business Mathematics, RWTH Aachen University
- 12/2022-07/2023 Member of the Search Committee W1 *Uncertainty Quantification*, RWTH Aachen University

Awards

- 10/2018 Werner-Oettli-Award for one of the best Master's theses in 2017/2018
- 06/2011 DMV-Abiturpreis

Engagement in the scientific community

- Network Coordination of a German-African network in the field of optimization and control
 ○ Continuation of the network that emerged from the *Recent Trends in Optimization and Control* workshops
 ○ Organization of online talks
 ○ Management of the email distribution list with over 60 members
- Guest editor **Special Issue: Nonlocal conservation laws, Networks and Heterogeneous Media**, <http://www.aimspress.com/nhm/article/6300/special-articles>
- 20 Reviews for
- | | |
|--------------------------------------|-----------------------------------|
| ○ Adv. Contin. Discrete Models | ○ Appl. Math. Lett. |
| ○ Bull. Braz. Math. Soc. | ○ Comput. Chem. Eng. |
| ○ Discrete Contin. Dyn. Syst. Ser. B | ○ ESAIM Math. Model. Numer. Anal. |
| ○ IMA J. Numer. Anal. | ○ Kinet. Relat. Models |
| ○ Math. Comput. Simulation | ○ Math. Biosci. Eng. |
| ○ Netw. Heterog. Media | ○ Optim. Eng. |
| ○ SEMA SIMAI Springer Series | ○ SIAM J. Appl. Dyn. Syst |
| ○ SIAM J. Appl. Math. | ○ Z. Angew. Math. Phys. |

Memberships

- since 07/2023 DFG-SPP 2410: *Hyperbolic Balance Laws in Fluid Mechanics: Complexity, Scales, Randomness*
- since 03/2023 Deutscher Hochschulverband
- 01/2023-12/2023 Society for Industrial and Applied Mathematics (SIAM)
- 01/2022-07/2023 DFG-SPP 1962: *Non-smooth and Complementarity-based Distributed Parameter Systems: Simulation and Hierarchical Optimization*

Publications

Articles in peer-reviewed journals

1. A. Beckers, J. Friedrich
Monotone-based numerical schemes for two-dimensional systems on nonlocal conservation laws
Adv. Contin. Discrete Models, 71 DOI:10.1186/s13662-026-04121-z, 2026
2. S. Sudha, J. Friedrich, S. Rathan
Convergence of the non-staggered Nessyahu-Tadmor scheme for coupled systems of one-dimensional nonlocal balance laws
IMA J. Numer. Anal., drag047, DOI:10.1093/imanum/drag047, 2026
3. J. Friedrich, S. Schraven, F. Kiessling, M. Herty
Source identification in bioluminescence tomography by consensus-based optimization
Optics Express, 33(16): 33312-33329, DOI:10.1364/OE.546936, 2025
4. F. Chiarello, J. Friedrich, S. Göttlich
A non-local traffic flow model for 1-to-1 junctions with buffer
Netw. Heterog. Media, 19 (1), 405-429, DOI: 10.3934/nhm.2024018, 2024
5. J. Friedrich, S. Göttlich, A. Keimer, L. Pflug
Conservation laws with nonlocal velocity - the singular limit problem
SIAM J. Appl. Math., 84 (2), 497-522, DOI: 10.1137/22M1530471, 2024
6. J. Friedrich, S. Göttlich, M. Herty
Lyapunov stabilization for nonlocal traffic flow models
SIAM J. Control Optim., 61 (5), 2849-2875, DOI: 10.1137/22M152181X, 2023
7. J. Friedrich, S. Sudha, S. Rathan
Numerical schemes for a class of nonlocal conservation laws: a general approach
Netw. Heterog. Media, 18 (3), 1335-1354, DOI: 10.3934/nhm.2023058, 2023
8. J. Friedrich, S. Göttlich, A. Uphoff
Conservation laws with discontinuous flux function on networks: a splitting algorithm
Netw. Heterog. Media, 18 (1), 1-28, DOI: 10.3934/nhm.2023001, 2023
9. A. Bayen, J. Friedrich, A. Keimer, L. Pflug, T. Veeravalli
Modeling multilane traffic with moving obstacles by nonlocal balance laws
SIAM J. Appl. Dyn. Syst., 21 (2), 1495-1538, DOI: 10.1137/20M1366654, 2022
10. J. Friedrich, S. Göttlich, M. Osztfalk
Network models for nonlocal traffic flow
ESAIM Math. Model. Numer. Anal., 56, 213-235, DOI: 10.1051/m2an/2022002, 2022
11. J. Friedrich, E. Rossi, S. Göttlich
Nonlocal approaches for multilane traffic models
Commun. Math. Sci., 19, 2291-2317, DOI: 10.4310/CMS.2021.v19.n8.a10, 2021
12. F. Chiarello, J. Friedrich, P. Goatin, S. Göttlich
Micro-Macro limit of a non-local generalized Aw-Rascle type model
SIAM J. Appl. Math., 80, 1841-1861, DOI: 10.1137/20m1313337, 2020

13. F. Chiarello, J. Friedrich, P. Goatin, S. Göttlich, O. Kolb
A non-local traffic flow model for 1-to-1 junctions
European J. Appl. Math., 31, 1029-1049, DOI: [10.1017/s095679251900038x](https://doi.org/10.1017/s095679251900038x), 2020
14. J. Friedrich, O. Kolb
Maximum principle satisfying CWENO schemes for nonlocal conservation laws
SIAM J. Sci. Comput., 41, A973-A988, DOI: [10.1137/18m1175586](https://doi.org/10.1137/18m1175586), 2019
15. J. Friedrich, O. Kolb, S. Göttlich
A Godunov type scheme for a class of LWR traffic flow models with non-local flux
Netw. Heterog. Media, 13, pp. 531-547, DOI: [10.3934/nhm.2018024](https://doi.org/10.3934/nhm.2018024), 2018

Peer-reviewed conference proceedings and book chapters

1. J. Friedrich, S. Rathan, S. Sudha
A note on the central-upwind scheme for nonlocal conservation laws
accepted to SEMA SIMAI Springer Series: Advances in Numerical Methods for Hyperbolic Balance Laws and Related Problems, [arxiv:2512.01344](https://arxiv.org/abs/2512.01344), to appear 2026
2. M. Banda, J. Friedrich, M. Herty
Boundary stabilization with restricted observability
accepted to Hyperbolic Problems: Theory, Numerics, Applications. HYP 2024, [arxiv:2501.15906](https://arxiv.org/abs/2501.15906), to appear 2026
3. M. Banda, J. Friedrich, S. Göttlich, M. Herty
Multi-scale control concepts for transport dominated problems
accepted to Non-Smooth and Complementarity-Based Distributed Parameter Systems Simulation and Hierarchical Optimization, Part II, to appear 2026
4. J. Friedrich, S. Göttlich, A. Keimer, L. Pflug
Conservation laws with nonlocality in density and velocity and their applicability in traffic flow modelling
Hyperbolic Problems: Theory, Numerics, Applications. Volume II. HYP 2022, DOI: [10.1007/978-3-031-55264-9_30](https://doi.org/10.1007/978-3-031-55264-9_30), 2024
5. J. Friedrich
Lyapunov stabilization of a nonlocal LWR traffic flow model
PAMM. Proc. Appl. Math. Mech., DOI: [10.1002/pamm.202200084](https://doi.org/10.1002/pamm.202200084), 2023

Preprints

- A. Beckers, J. Friedrich
A third-order maximum-principle-preserving CWENO scheme for two-dimensional nonlocal conservation laws
[arXiv:2607.19953](https://arxiv.org/abs/2607.19953), 2026
- J. Friedrich, M. Herty, C. Nocita
Control of conservation laws in the nonlocal-to-Local limit
[arxiv:2510.00677](https://arxiv.org/abs/2510.00677), 2025

Academic theses

- J. Friedrich - Traffic flow models with nonlocal velocity, Ph.D Thesis, University of Mannheim, Verlag Dr. Hut, ISBN 978-3-8439-4903-3, 2021, [PDF \(Researchgate\)](#)
- J. Friedrich - Network models and numerical methods for traffic flow with non-local flux terms, Master-Thesis, University of Mannheim, 2017
- J. Friedrich - Szenario-Aggregation für Risikomaße, Bachelor-Thesis (in German), University of Mannheim, 2015

Conferences and Workshops

- as Organizer **Recent Trends in Optimization and Control**, *Organizing & Scientific Committee*, Follow-up Workshop, Dakar, Senegal, 10/2024
<https://www.igpm.rwth-aachen.de/workshop/optcon2024>
- Recent Trends in Optimization and Control**, *Organizing & Scientific Committee*, Short Course und Workshop, Pretoria, South Africa, 09/2023
<https://www.igpm.rwth-aachen.de/workshop/optcon2023>
- SIAM PD 2022**, *Co-Chair*, Minisymposium: Nonlocal conservation laws, virtual, 03/2022
- DMV-ÖMG Annual Meeting 2021**, *Co-Chair*, Minisymposium 13: Nonlocal conservation laws, virtual, 09/2021
- 13th International Conference on Monte Carlo Methods and Applications**, *local Organizer*, virtual, 08/2021
- Talks **HYP 2026**, *A nonlocal transport model for human blood cell dynamics*, Stuttgart, 05/2026
- Oberseminar: Dynamics (TU Munich)**, *Invited speaker: An introduction to nonlocal conservation laws - Theory and numerics*, Munich, 04/2026
- Oberseminar: Numerical methods in CSE (TU Munich)**, *Invited speaker: Conservation laws with space-dependent nonlocalities*, Munich, 07/2025
- NumHyp 2025**, *Maximum principle satisfying CWENO schemes for multidimensional nonlocal conservation laws*, Darmstadt, 06/2025
- DataHyKing Workshop**, *Invited speaker: High-order schemes for nonlocal balance laws*, Aachen, 04/2025
- SPP2410: Annual Status Meeting**, *Invited speaker: Balance laws with space-dependent nonlocalities: modeling, simulation, UQs*, Darmstadt, 03/2025
- Monthly Lecture Series, Mahindra University**, *Invited speaker: Numerical schemes for conservation laws with space-dependent nonlocalities*, Hyderabad, Indien, 02/2025
- Numerical Aspects of Hyperbolic Balance Laws and Related Problems**, *Invited speaker: Nonlocal balance laws: Numerical schemes and applications to traffic flow*, Ferrara, Italy, 12/2024
- Recent Trends in Optimization and Control: Online Lectures**, *Source identification by consensus-based optimization*, virtual, 11/2024
- Control and Optimization in the Age of Data**, *Invited speaker: Source identification via a consensus-based optimization algorithm using different moment hierarchies*, Bayreuth, 09/2024
- Nonlocal Modelling in Fluidmechanical Applications**, *Invited speaker: Numerical schemes for conservation laws with space-dependent nonlocalities*, Mannheim, 09/2024
- HYP 2024**, *Lyapunov Stabilization for Nonlocal Traffic Flow Models*, Shanghai, China, 07/2024
financially supported by HYP 2024
- Seminar Talk at University of Twente**, *Source identification using different moment hierarchies and consensus-based optimization*, Enschede, Netherlands, 06/2024
- Recent Trends in Optimization and Control**, *Plenary talk: Cell tracking for the radiative transfer equation*, Pretoria, South Africa, 09/2023

- 16th Hirschegg Workshop on Conservation Laws**, *Traffic flow models with nonlocal velocity: The singular limit problem*, Hirschegg, Austria, 09/2023
- SIMAI 2023**, *Invited speaker: Cell tracking for the radiative transfer equation*, Matera, Italy, 08/2023
- SIAM OP 2023**, *Invited speaker: Cell tracking using uncertainty quantification for the radiative transfer equation*, Seattle, USA, 05/2023
- SIAM CSE 2023**, *Invited speaker: Aspects of nonlocal traffic flow modeling*, Amsterdam, Netherlands, 02/2023
- SPP 1962 Annual Meeting 2022**, *Lyapunov stabilization for nonlocal traffic flow models*, Berlin, Germany, 10/2022
- GAMM 2022**, *Lyapunov stabilization for nonlocal traffic flow models*, Aachen, Germany, 08/2022
- HYP 2022**, *Traffic flow models with nonlocal velocity: The singular limit problem*, Malaga, Spain, 06/2022
- Seminar Talk at FAU**, *Lyapunov stabilization of nonlocal traffic flow models*, Erlangen, Germany, 06/2022
<https://www.math.fau.de/events/vortrag-dr-jan-friedrich/>
- SIAM PD 2022**, *Nonlocal approaches for multilane traffic models*, virtual, 03/2022
- DMV-ÖMG Annual Meeting 2021**, *Network models for nonlocal traffic flow*, virtual, 09/2021
- GAMM Workshop on Numerical Analysis**, *Maximum principle satisfying CWENO schemes for non-local conservation laws*, Augsburg, Germany, 10/2018
- IFIP TC 7 Conference on System Modelling and Optimization**, *Invited speaker: Traffic flow models with non-local flux and approaches for network models*, Essen, Germany, 06/2018
- DMV Student Conference**, *Network Models and Numerical Methods for Traffic Flow with Non-Local Flux Terms*, Paderborn, Germany, 03/2018
- Spring School 2018: From Particle Dynamics to Gradient Flows**, *A Godunov type scheme for a class of scalar conservation laws with non-local flux*, Kaiserslautern, Germany, 02/2018
- Other **ProLehre Onboarding**, *Elective Module: Designing Courses*, Munich, Germany, 09/2025
- SPP 1962 Young Researchers' Workshop on Deep Learning**, Essen, Germany, 03/2023
 financed by DFG-SPP 1962
- Manage your biases: How to outsmart your own unconscious bias**, *Participation in Unconscious Bias Training from fisch & friends international*, online, 03/2023
 financed by DFG-SPP 1962
- Klartext Workshop on Scientific Communication 2022/2023**, *Participation in the workshop from the Klaus Tschira foundation in cooperation with the National Institute for Scientific Communication*, Heidelberg, Germany, 01/2023
 financed by the Klaus Tschira foundation
- Normandy Meeting on Theoretical and Numerical Aspects Of PDEs**, *Poster: A one-to-one junction for a LWR traffic flow model with non-local flux*, Rouen, France, 11/2018
 financially supported by Région Normandie and COMUE Normandie Université

Supervision and mentoring

since 06/2024 **Anika Beckers**, *Numerical methods for nonlocal balance laws in multiple dimensions*, PhD student of Prof. Dr. Michael Herty, RWTH Aachen University
financed by DFG-SPP 2410

Teaching

- Lectures **Mathematics III (Numerics for civil engineers)**, *Technical University of Munich*, Winter 2026
Case studies optimization, *Technical University of Munich*, Summer 2026
Applications of scalar conservation laws, *RWTH Aachen University*, Winter 2023
Numerical methods for ordinary differential equations, *University of Mannheim*, Winter 2021
- Courses **Applications of scalar hyperbolic conservation laws**, *Indian Institute of Petroleum and Energy*, March 2024, PhD course
10 lectures in English (online), ≈ 40 participants
Introduction to MATLAB, *University of Mannheim*, Winter 2019 and 2020 (online), Introductory course
part of the lecture Numerical mathematics
- Exercise classes **Mathematics III (Numerics for civil engineers)**, *Technical University of Munich*, Winter 2026
Nonlinear Optimization: Advanced, *Technical University of Munich*, Winter 2025
Mathematics I (for civil engineers), *RWTH Aachen University*, Winter 2022–2024
Mathematics II (for civil engineers), *RWTH Aachen University*, Summer 2022–2025
Numerical methods for partial differential equations, *University of Mannheim*, Summer 2019 and 2020–2021 (online)
Applications of scalar conservation laws, *University of Mannheim*, Winter 2019 and 2020 (online)
Analysis for business informatics, *University of Mannheim*, Summer 2019
Numerical methods for ordinary differential equations, *University of Mannheim*, Winter 2018
Numerical mathematics, *University of Mannheim*, Summer 2015–2016 and 2018
Linear Optimization, *University of Mannheim*, Winter 2017
Analysis 1, *University of Mannheim*, Winter 2015
- Seminars **Modeling and simulation**, *University of Mannheim*, Summer 2018 and 2019